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A prospective cohort study evaluating the outcome of root canal retreatment in symptomatic mandibular first molars with periapical lesions

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INTRODUCTION

The clinical decision-making process for the management of the post-treatment endodontic disease following root canal treatment is challenging. The main reasons for disease following either root canal treatment or root canal retreatment are the presence of intraradicular or extraradicular infection which could not be completely eliminated (Haapasalo et al., 2011; Nair et al., 2005) even in teeth that were apparently well treated (Sjögren et al., 1990) possibly due to infection remaining in lateral canals, isthmuses and dentinal tubules (Carr et al., 2009; Ricucci et al., 2009). Root canal retreatment and endodontic surgery are the commonly preferred treatment approaches to promote long-term tooth survival in cases with post-treatment disease.

The data obtained from prospective clinical studies are crucial for evidence-based dentistry; however, most of the studies related to root canal retreatment have been retrospective in nature (Çalışkan, 2005; Curtis et al., 2018; Dammaschke et al., 2003; Eyuboglu et al., 2017; Nieuwenhuysen et al., 1994; Olcay et al., 2019). In the literature, 16 prospective studies have been conducted to address the outcome of root canal retreatment; however, these studies have major inadequacies with respect to standardization, such as inclusion of both symptomatic and asymptomatic teeth, different tooth types with and without periapical lesions (Al-Nuaimi et al., 2017; Danin et al., 1996; Davies et al., 2016; De chevigny et al., 2008; Farzaneh et al., 2004; Gorni & Gagliani, 2004; He et al., 2017; Kvist & Reit, 1999; Mente et al., 2015; Metska et al., 2013; Ng et al., 2011a,b; Riis et al., 2018; Sundqvist et al., 1998; Zandi et al., 2019; Zhang et al., 2021). Although the management of pain in previously treated teeth is crucial and the intraradicular microbial flora which is the main reason for the failure of primary root canal treatment differs in symptomatic and asymptomatic teeth, most of the studies fail to provide information on the outcome of the root canal retreatment in cases with both clinical symptoms and periapical lesions (De chevigny et al., 2008; Davies et al., 2016; Farzaneh et al., 2004; He et al., 2017; Kvist & Reit, 1999; Mente et al., 2015; Ng et al., 2011a; Riis et al., 2018).

Therefore, the aim of this prospective study was to evaluate the influence of various predictors on the outcome of root canal retreatment in symptomatic mandibular first molars with periapical lesions after a minimum of 2 years.

MATERIALS AND METHODS

Study design and patient population

This study followed the STROBE checklist and was conducted after obtaining ethical approval from the Local Ethical Committee (no. 15-6.1/8) in accordance with the principles of the Helsinki (version 2008). Amongst the 420 mandibular molars from patients who were referred for root canal retreatment between February 2016 and December 2018, 120 root canal treated symptomatic mandibular first molars with periapical lesions were included. All patients were informed about the potential complications of the root canal retreatment procedure, number of visits and follow-up examinations before their participation to this study and written informed consents were obtained. This prospective cohort study has been written according to the Preferred Reporting Items for Observational studies in Endodontics 2020 guidelines (Nagendrababu et al., 2020). Figure 1 describes the design of the study and flow of patients.

Inclusion criteria

Patients selection and inclusion criteria were as follows: patients with noncontributory medical history; previously root canal treated mature mandibular first molars with clinical signs and symptoms, including preoperative pain and tenderness to percussion; at least one root with periapical lesions which were scored as PAI 4 and 5 using periapical index scale (PAI; Ørstavik et al., 1986); a periapical lesion whose largest initial diameters ranged between 3 and 10 mm on preoperative digital radiographs that were taken using paralleling technique with a film holder (Kerr Dental) and were diagnosed as symptomatic apical periodontitis or acute apical abscess with localized intraoral swelling.

Exclusion criteria

Exclusion criteria were as follows: patients younger than 18 years of age; extra-oral swelling; antibiotic intake within 2 weeks prior to the beginning of the root canal retreatment; second and third molars; teeth with asymptomatic apical periodontitis; two adjacent teeth which required root canal retreatment; teeth with advanced

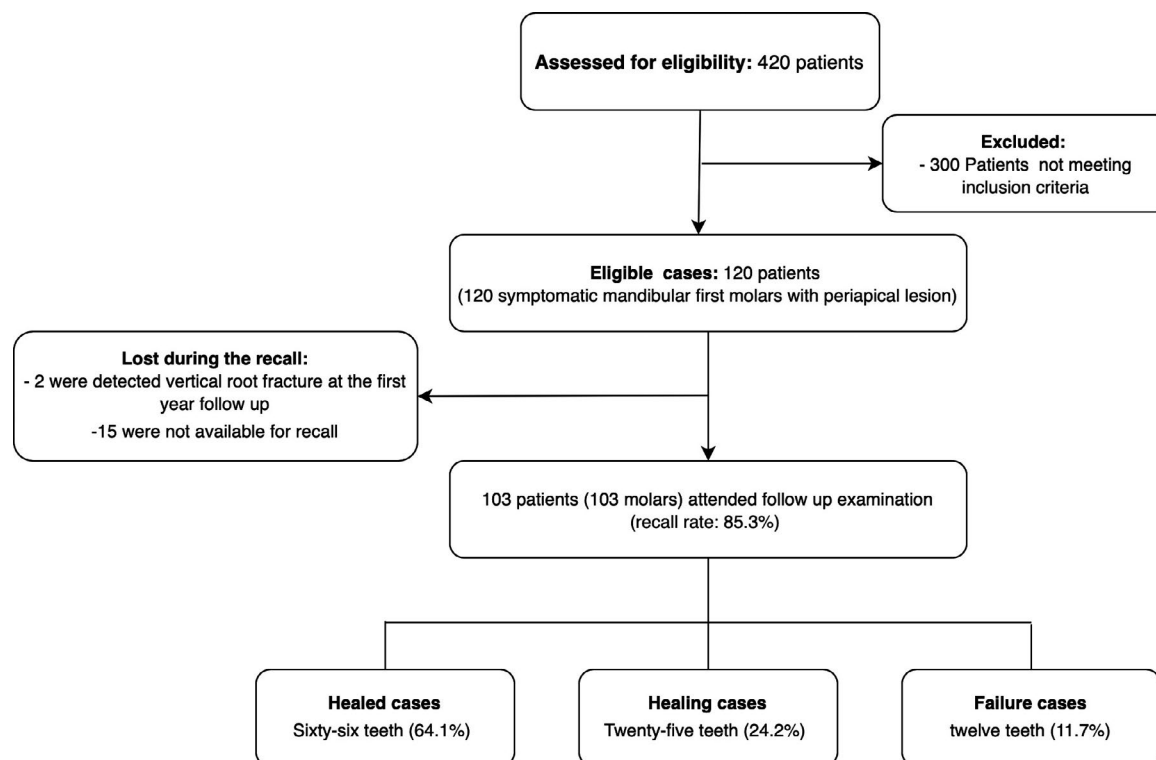


FIGURE 1 Flow chart of participants and recall rate.

periodontal disease with periodontal pocket depth >5 mm and/or loss of attachment of ≥ 3 mm, nonrestorable loss of tooth structure, vertical root fracture, external/internal root resorption and presence of complication related to the primary root canal treatment, such as broken instrument or perforation.

Data collection and treatment protocol

The preoperative predictors, including patients' age and gender, soft tissue tenderness (present/absent), intraoral swelling (present/absent), periapical lesion size (3–10 mm) and PAI score (4 or 5), apical level (short/adequate/overfill) and density of root canal filling (poor: presence of visible voids within gutta-percha or between gutta-percha and the root canal walls; good: homogeneous root canal filling and no visible voids; unfilled: no gutta-percha inside the canal) were recorded before root canal retreatment. All root canal retreatments were performed with the aid of magnifying loupes ($\times 3.5$ magnification) (Eye-Mag Pro, Carl Zeiss) by an endodontist with 8 years of experience. A dental operating microscope (OPMI Pico; Carl Zeiss Surgical GmbH) was also used in cases where the endodontist suspected the presence of an extra canal or to locate root canal orifices.

After performing local anaesthesia (Ultracaine DS; Aventis Pharma), occlusal trauma which was detected

by using articulating paper was eliminated and rubber dam isolation was provided. Following removal of defective restorations and caries, an access cavity was prepared to establish direct access to the canal orifices. Pre-existent root canal filling material was removed mainly using ProTaper Universal Retreatment files (PTU-R; Dentsply Sirona) and if necessary Hedström hand files without using solvent. When PTU-R instruments could not reach working length (WL) and resistance was met with a tactile tacky feeling by the instruments, small K-files (sizes 06, 08, 10; Dentsply Sirona) were used to penetrate and negotiate the root canals. After negotiation, rotary instruments were utilized to reach the entire WL. In cases when negotiation could be accomplished, apical patency was checked and confirmed using a Raypex VI electronic apex locator (VDW) in 98 teeth. However, in five molars establishment of apical patency could not be achieved. Thus, the reshaping of the canals was completed to the maximum length that was reached in these teeth. During the removal of the gutta-percha, 2-mL sodium hypochlorite (NaOCl; Merck) and 2-mL ethylenediaminetetraacetic acid (EDTA) (Merck) were used with a 30-G side-vented irrigation needle (Sybron Endo Corp) after the use of each PTU-R files. When gutta-percha was removed from the canals, WL was determined using an apex locator (Raypex VI; VDW) and was confirmed with digital periapical radiographs.

Cases with exudate

Afterwards, in all cases with or without intraoral swelling, an attempt was made to check whether there was an active exudate draining through the root canal. In teeth with active exudate drainage through the root canal that did not cease during the debridement, reshaping of the root canals was performed using ProTaper Universal System (PTU; Dentsply Sirona) until the instruments reached to the WL, and no remaining root canal filling material or decalcified dentine remnants were observed on the instrument flutes whilst examining with a magnifying loupe ($\times 3.5$). The root canal preparation was completed with PTU F2 (size 25, .08 taper) or F3 (size 30, .09 taper) instruments in mesial canals and F3 (size 30, .09 taper), F4 (size 40, .06 taper) or F5 (size 50, .05 taper) files in the distal canals depending on the canal size and anatomy. However, an intracanal dressing of calcium hydroxide (CH; Merck) was not placed in the root canals in the first appointment because of the active exudate drainage. Access cavities were filled with glass-ionomer cement (Ketac-Molar; 3 M ESPE). No antibiotics were prescribed. At the second appointment, the persistence of the swelling was checked. In order to evaluate the exudate, sterile paper points were inserted into the root canals to assess for moisture in the canal. When no moisture was observed on the paper points at the second or third appointments, the root canals were irrigated with 10 mL of 2.5% NaOCl per canal. A paste of CH mixed with sterile saline was applied to the canals for 7–10 days using a lentulo spiral fillers (Dentsply Sirona) and access cavities were restored with glass-ionomer cement.

At the last appointment, the CH was removed using a combination of hand files, 17% EDTA and 2.5% NaOCl. Final irrigation was performed with 5 mL of 17% EDTA, 5 mL of 2.5% NaOCl and manual dynamic activation with sterile gutta-percha was performed during the final irrigation using both NaOCl and EDTA for 1 min. Afterwards, root canals were rinsed with 5-mL distilled water and dried with paper points, root canals were irrigated with 5 mL of 2% chlorhexidine (CHX, Klorhex; Drogosan) and it was allowed to contact with the root canal walls for up to 2 min. Afterwards, the teeth were subsequently root filled with modified single cone technique similar to Krug et al. (2017). Following the placement of matching master PTU gutta-percha points (Dentsply, Sirona) associated with the finishing file for each root canal using AH Plus sealer (Dentsply DeTrey), the area between the PTU gutta-percha and canal periphery was checked with a spreader. If there was a gap, accessory cones were inserted into the root canals until the spreader did not reach deeper than 2 mm below the canal orifices. The teeth were restored with either direct composite resin (Clearfil Majesty

Esthetic; Kuraray Medical) or a core build-up and single porcelain crown.

Cases without exudate

In cases where exudate ceased during debridement, root canals were re-shaped using PTU and irrigation was performed between each file with 2.5% NaOCl by delivering a total volume of 10 mL of irrigant per canal. After completion of the root canal re-shaping, CH was applied to the canals for 7–10 days and access cavities were restored with glass-ionomer cement. At the second appointment, final irrigation, root filling and restoration procedures were performed exactly as in the exudate group.

For each group, the intraoperative predictors, such as the active exudate drainage through the root canal (present/absent), apical level and density of root canal filling after root canal retreatment (good/poor), complications such as sealer and/or gutta-percha extrusion, broken instrument, perforations and flare up and restoration type (single crown/composite) were recorded during the treatment visits.

Assessment of outcome and follow-ups

Patients were recalled for follow-up examinations after the root canal retreatments were completed at 3, 6, 9 and 12 months for the first year, every 6 months for the second year and every year thereafter until the end of the study. All patients were followed up for clinical and radiographic examinations for a minimum 2 years up to 4 years. At the follow-up appointments, clinical examination was performed, including the presence of pain, tenderness to percussion and palpation, sinus tract or swelling and the mobility. Additionally, the quality of the restorations in terms of secondary caries, degradation in marginal adaptation, evidence of discrepancy and decementation was also evaluated. For the radiographic assessment, both pre- and postoperative radiographs were taken using a long-cone paralleling technique with a film holder under standard exposure conditions. In order to improve the reproduction of anatomy, the X-ray beam was placed perpendicular to the phosphor plate by centring and aligning on the aiming ring of the film holder. Additionally, the position of the bite-block of the film holder was checked by visualizing the preoperative radiograph for each patient before taking the follow-up radiographs. Two independent experienced endodontists, who had been calibrated on how to use the PAI scoring system (Ørstavik et al., 1986) on a series of 100 digital periapical radiographic images unrelated to the present study, independently scored the initial and final

radiographs to prevent any bias. In case of disagreement, they scored the radiographs together until agreement was reached. When the observers did not agree on a PAI score, the worst score was recorded. For the teeth with two separate periapical lesion on different root, the worst PAI score was considered as the final PAI score of the teeth.

The outcome of root canal retreatment was evaluated using three categories based on the modified PAI scoring system (Holden et al., 2008) as follows:

Healed: The tooth had no clinical signs or symptoms, and the radiographs revealed complete resolution of periapical lesion as scoring PAI 1 or 2.

Healing: The tooth had no clinical signs or symptoms, accompanied by a reduction in the lesion size and PAI score of 3 or 4 at follow-up.

Nonhealing: The tooth was symptomatic or asymptomatic, and the size of the periapical lesion was unchanged or increased and PAI score 4 or 5 or previously normal periodontal ligament space had developed into a periapical lesion compared to the preoperative radiographs.

Nonhealing was considered as a “failure” and the combination of the healing and healed cases considered as a “success”.

were diagnosed at the first year follow-up, and 15 were not available for recall. Ultimately, a total of 103 mandibular first molars were evaluated depending on the clinical and radiographic criteria at the last follow-up (Figure 1). An overall recall rate was 85%. All of the root canal retreatments were completed in two to four visits. Of the included 103 teeth, root canal retreatment was performed in two appointments in 75 teeth without active exudate drainage, and in three appointments in 20 teeth with active exudate drainage. Root canal retreatments were completed in four appointments in the remaining eight teeth, four of which had flare-ups and the other four had exudate drainage up to the third appointment.

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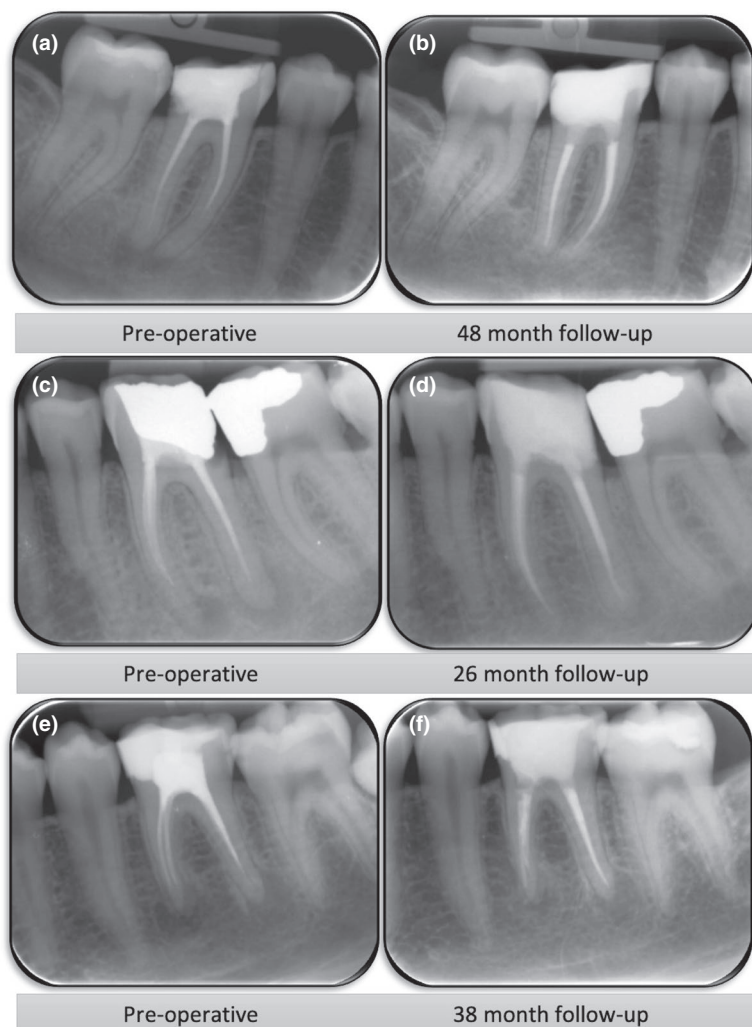


FIGURE 2 Representative periapical radiographs of (a and b) healed, (c and d) healing and (e and f) failure cases. (a) Preoperative and (b) 48-month follow-up radiographs showing complete resolution of the periapical radiolucency; case was considered “healed”. (c) Preoperative and (d) 26-month follow-up radiographs showing reduced periapical radiolucency; case was considered “healing”. (e) Preoperative and (f) 38-month follow-up radiographs showing that mesial root reduced periapical radiolucency; however, distal root unchanged in size of the periapical lesion; case was considered as “failure”.

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CONFLICT OF INTEREST

The authors have stated explicitly that there are no conflicts of interest in connection with this article.

ETHICAL APPROVAL

This study was conducted after obtaining ethical approval from the Local Ethical Committee (no. 15-6.1/8) in accordance with the principles of the Helsinki (version 2008).

AUTHOR CONTRIBUTIONS

The authors confirm contribution to the paper as follows: study conception and design (Serefoglu B, Çalışkan MK), data collection (Serefoglu B, Kandemir Demirci G, Miçooğulları Kurt S), analysis and interpretation of results (Serefoglu B, Kandemir Demirci G, Miçooğulları Kurt S, Kaval ME, Çalışkan MK), draft manuscript preparation (Serefoglu B, Kandemir Demirci G, Miçooğulları Kurt S, Kaval ME, Çalışkan MK). All authors reviewed the results and approved the final version of the manuscript.

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